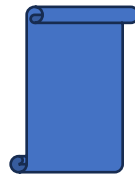




📌 PROJECT OVERVIEW

This project presents an **end-to-end retail sales analytics solution**, combining **MySQL for data preparation, transformation, and business analysis** with **Power BI for interactive visualization and storytelling**. The objective was to transform raw retail data into meaningful business insights across **sales, profitability, customers, products, stores, regions, and payment behavior**.

The project follows a practical analytics workflow - starting with understanding the underlying data structure and correcting data types, continuing with **SQL-based exploration and business-question analysis**, and ultimately building an interactive **four-page Power BI dashboard**. The final solution enables stakeholders to quickly understand **what is driving revenue and profit, which products and stores are performing best, who the most valuable customers are, and where opportunities for improvement exist**. 📊



🎯 THE PROBLEM STATEMENT:

Retail businesses generate large volumes of transactional data, but raw transaction records alone do not provide an immediate understanding of **business performance or customer behavior**. The goal of this project was to develop a centralized analytical solution that could answer questions such as:

- How much revenue and profit is the business generating?
- How is sales performance changing over time?
- Which products and categories contribute the most revenue?
- Which stores and regions are performing best?
- Who are the highest-value customers?
- What customer demographics dominate the business?
- Which payment methods are most frequently used?
- How many customers are new versus returning?
- Which customers have become inactive?

The ultimate objective was to move from **raw transactional data** → **structured analysis** → **interactive business intelligence** → **actionable decisions**. 🔄



DATASET DETAILS

The dataset represents a **multi-store retail chain** and consists of four interconnected tables:

Customer Data : 200 customer records and 7 fields

- Customer ID
- First Name
- Last Name
- Gender
- Birth Date
- City
- Join Date

This table supports analysis of **customer demographics, age groups, gender distribution, customer acquisition and customer lifetime activity.**

Product Data : 50 rows and 6 fields

- Product ID
- Product Name
- Category
- Subcategory
- Unit Price
- Cost Price

The product table enables analysis of **product-level revenue, profitability, purchasing patterns, categories, and subcategories.**

Store Data : 5 stores and 4 fields.

- Store ID
- Store Name
- City
- Region

This information supports **store-level and regional performance analysis.**

Transaction Data : 5,000 transactions and 8 fields

- Transaction ID
- Date
- Customer ID
- Product ID

- Store ID
- Quantity
- Discount
- Payment Method

The transaction table acts as the primary fact table and connects customers, products, and stores to actual sales activity.



DATA EXPLORATION AND CLEANING

Before loading the data into Power BI, I first focused on **understanding the structure, quality and characteristics of the underlying dataset**. The data was reviewed in MySQL to identify the available fields, relationships, data types, and potential analytical requirements.

Key Data Preparation Activities

- **Reviewed and corrected data types**, particularly date-related fields.
- Verified numerical fields such as **quantity, price, cost and discount**.
- Ensured geographic attributes such as **city and region** could be appropriately interpreted for mapping.
- Examined categorical fields such as **gender, category, subcategory, store and payment method**.
- Checked relationships between **customers, products, stores, and transactions**.
- Used SQL joins and aggregations to create analysis-ready datasets.
- Developed **20+ SQL business queries** to explore the data before visualization.

I also paid attention to how Power BI interpreted each field - distinguishing between **dates, numerical values, categorical fields, geographic fields and text attributes** before building visualizations. This step helped ensure that the dashboard was not simply visually attractive, but also **built on correctly interpreted and structured data**.   



✚ WHAT'S INSIDE THE DASHBOARD

The Power BI solution consists of **four interactive analytical pages**, each designed around a different business perspective:

💰 1. Sales Overview

Provides a high-level view of **revenue, profitability, orders, payment methods, stores and sales trends**.

🛒 2. Product Analysis

Explores **product performance, categories, subcategories, revenue, profitability and purchasing behavior**.

👤 3. Customer Analysis

Focuses on **customer demographics, spending, gender, age groups, acquisition trends and customer value**.

📱 4. Store Performance

Evaluates **store, country, and regional performance**, including revenue, profit, orders and quantity.

Interactive slicers allow users to dynamically filter the analysis by **Country, Gender, Category and Date**, making it easier to move from an overall business view to a more focused analysis



🧰 TECHNOLOGY STACK

Technology	Purpose
📱 MySQL	Data exploration, cleaning, transformation & SQL analysis
📊 Power BI	Dashboard development & interactive visualization
📈 DAX	Measures, KPIs, time intelligence & calculations
🔗 Data Modeling	Connecting customers, products, stores & transactions
💻 SQL	Joins, aggregations, filtering & business-question analysis



DAX Formulas & New Measures

DAX was used to convert raw transactional information into **business-focused KPIs and analytical measures**. Key measures included calculations for:

- Total Revenue
- Total Profit
- Previous Year Revenue
- Previous Year Profit
- Revenue Growth %
- Profit Growth %
- Average Order Value
- Total Orders
- Average Spend per Customer
- Average Spend per Transaction
- Customer Counts
- New Customers
- Product and Store performance metrics

For example, time-intelligence measures were used to compare **current-year performance against the previous year**, allowing the dashboard to communicate not only how much the business generated but also how performance changed over time.



VISUALS UTILIZED TO ANALYZE THE DATA

The dashboard combines different visualization techniques to make the analysis accessible from both **high-level and detailed perspectives**.

Sales Analysis

- KPI Cards
- Gauge Chart
- Donut Chart
- Combo Chart

- Line Chart

Product Analysis

- Bar Chart
- Scatter Plot
- Decomposition Tree
- Treemap

Customer Analysis

- Line Chart
- Donut Chart
- Bar Chart
- Combo Chart
- Treemap

Store & Geographic Analysis

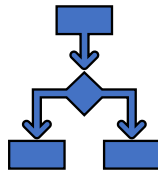
- Bar Chart
- Line Chart
- Matrix
- Geographic Map
- **Orthographic Shape Map** 🌐 : The **Orthographic Shape Map** was particularly useful for geographic storytelling and was also an opportunity to experiment with a Power BI feature I had recently learned and incorporate it into a practical analytics project



UI DESIGN AND THEME COLORS

The dashboard was designed with a **clean, professional, and user-friendly interface**, using **light blue as the primary theme color** to create a calm and consistent visual experience across the dashboard. To explore different visual styles and improve emphasis, I also incorporated **orange as an accent color** across the individual dashboard pages. The combination of **light**

blue and orange helped create visual contrast, highlight important KPIs and insights, and make key elements easier to distinguish at a glance. The overall design focused on maintaining **consistency, readability, visual hierarchy, and an engaging user experience** while keeping the dashboard professional and business-oriented. 🎨 📊



🔄 Steps Followed to Build the Dashboard

The project followed a structured **data-to-insight workflow**:

1 Understand the Dataset

Reviewed tables, fields, relationships and data types.

2 Prepare the Data in MySQL

Corrected data types and prepared structured datasets for analysis.

3 Explore the Data Using SQL

Created **20+ business-focused SQL queries** covering customers, products, stores, sales, profitability and trends.

4 Identify Business KPIs

Selected metrics such as **Revenue, Profit, Orders, AOV, Customer Spend, Growth and Quantity**.

5 Connect Data to Power BI

Imported the prepared data and established the necessary relationships.

6 Configure Data Categories

Assigned appropriate classifications such as **date, geographic, numerical and categorical fields**.

7 Develop DAX Measures

Created calculated measures for KPIs, comparisons, growth analysis and customer/product metrics.

8 Build Four Dashboard Pages

Designed dedicated pages for **Sales, Products, Customers and Stores**.

9 Add Interactivity

Implemented slicers for **Country, Gender, Category and Date**.

10 Validate & Refine

Cross-checked calculations and visual outputs to ensure consistency and meaningful storytelling.

1 1 Apply Visual Storytelling

Selected visualizations based on the analytical question rather than simply filling dashboard space.



💡 KEY INSIGHTS

The analysis revealed several notable business patterns:

💰 Revenue & Profitability

- **Total Revenue reached ₹15.4M**, compared with **₹9.96M in the previous year**.
- Revenue increased by approximately **54% YoY**.
- **Total Profit reached ₹5M**, compared with **₹3.14M in the previous year**.
- Average Order Value was approximately **₹3.1K**.

🌍 Geographic Performance

- **Russia generated the highest revenue** among the countries analyzed.
- The **East Region** emerged as the best-performing region.
- **MegaMart**, located in the East Region/Russia, was the strongest-performing store, generating approximately **₹3.16M in revenue and 1K orders**.

🛒 Product Performance

- **Accessories** was the leading category, generating approximately **₹2M in revenue across 556 orders**.
- **Road Clothing** was identified as the most purchased product, generating approximately **₹139K revenue from 27 purchases**.
- The product analysis highlighted differences between **revenue contribution and profitability**, helping identify products that may warrant closer attention.

Customer Behavior

- The dataset contained **200 customers**.
- **Male customers represented approximately 56.5%** of the customer base.
- Customers aged **50–60 represented the largest age group**.
- **Brittney** emerged as the highest-value customer, generating approximately **₹153K revenue across 50 orders**.
- **23 new customers** joined during the last six months analyzed.

Payment Behavior

- **Credit Card and Cash** were among the most popular payment methods, each accounting for roughly **25%** of transactions.
- Credit Card transactions alone represented approximately **25.68%** of revenue/transaction activity as reflected in the dashboard analysis.

These findings demonstrate that performance was not driven by one factor alone - **geography, product mix, customer value and store performance all contributed to the overall business picture**.



◆ THE STORYTELLING

 The dashboard was designed to tell a **progressive business story**.

It begins with the question:

How is the business performing overall?

The **Sales Overview** answers this through revenue, profit, orders, payment methods and sales trends.

From there, the analysis moves to:

What products are driving that performance?

The **Product Analysis** explores product, category, and subcategory contribution while comparing revenue and profitability.

The next question becomes:

Who are the customers behind the sales?

The **Customer Analysis** reveals customer demographics, spending patterns, acquisition trends, and high-value customers.

Finally: **Where is the business performing best?**

The **Store Performance** page connects sales performance with stores, regions, and countries.

This creates a natural flow:

Business Performance → Products → Customers → Stores & Geography → Strategic Decisions



⚠ CHALLENGES & HOW THEY WERE ADDRESSED

The dataset contained multiple interconnected tables, making it important to understand how customers, products, stores, and transactions related to each other.

Solution:

Reviewed the schema and relationships before beginning visualization and used SQL joins to create structured analytical outputs.

Challenge 2: Correct Data Interpretation

Fields such as dates, geographic attributes, numerical fields, and categorical dimensions need to be correctly interpreted by Power BI.

Solution:

Reviewed and adjusted data types and classifications before building the dashboard, including assigning appropriate geographic categories to location fields.

Challenge 3: Converting Business Questions into Visuals

Not every business question can be answered effectively using the same visualization.

Solution:

Selected visuals based on the analytical purpose — trends used line charts, comparisons used bar charts, contribution used donut/treemap visuals, relationships used scatter plots, and hierarchical exploration used a decomposition tree.

Challenge 4: Moving Beyond Raw Numbers

Simply displaying revenue and profit does not explain whether the business is improving.

Solution:

Created DAX measures for **previous-year comparisons, growth, averages, and customer/product KPIs**, allowing the dashboard to communicate performance changes rather than isolated numbers.

Challenge 5: Geographic Storytelling

Geographic performance can be difficult to communicate through tables alone.

Solution:

Experimented with Power BI's **Orthographic Shape Map** to make geographic patterns more visually intuitive



WHY THIS PROJECT MATTERS

- Retail analytics is ultimately about making better decisions from customer and transaction data.
- A business may know that it generated **₹15.4M in revenue**, but that number becomes much more valuable when decision-makers can understand:
- **Where did the revenue come from?**
 - Which products contributed to it?**
 - Which customers generated the most value?**
 - Which stores and regions are outperforming?**
 - How is performance changing compared with the previous year?**
- This project demonstrates how an analyst can move beyond **descriptive reporting** and create a structured BI solution that connects **data preparation, SQL analysis, data modeling, DAX, visualization, and business interpretation**.



STRATEGIC RECOMMENDATIONS

Based on the analysis, several strategic actions can be considered:

1. Strengthen High-Performing Stores

MegaMart and the East Region demonstrated strong performance. Management should identify the operational practices behind this success and evaluate whether they can be replicated across lower-performing stores.

2. Prioritize High-Performing Categories

With **Accessories leading category performance**, inventory planning and promotional strategies could prioritize products with strong demand while continuing to monitor profitability.

3. Develop High-Value Customer Strategies

Customers such as **Brittney**, who generate significantly higher revenue, represent valuable customer segments. Personalized offers, loyalty programs, and targeted promotions could help increase retention and lifetime value.

4. Focus on Customer Segmentation

The concentration of customers within the **50–60 age group** suggests an opportunity to develop targeted campaigns, while still exploring strategies to attract underrepresented customer segments.

5. Learn From Strong Geographic Markets

Since **Russia and the East Region** are leading contributors, the business should investigate the factors behind their success — including product mix, customer behavior, pricing, and store operations.

6. Optimize Payment Experience

With **Credit Card and Cash** being major payment methods, the business should continue supporting these options while monitoring emerging payment preferences to ensure a convenient customer experience.

7. Monitor Growth, Not Just Revenue

The **54% YoY revenue growth** is encouraging, but management should continuously monitor whether growth is accompanied by sustainable **profitability, customer retention, and healthy product margins**.

This project demonstrates a complete **Business Intelligence workflow**, starting with raw transactional data and ending with an interactive decision-support dashboard.

The combination of **MySQL + SQL + Data Modeling + DAX + Power BI** helped transform 5,000 transactions and supporting customer, product, and store information into a structured analytical solution.

More importantly, the project reinforced an important principle of analytics:

Good dashboards don't just display numbers - they help stakeholders understand what happened, why it happened, and where to act next.  